

1 Code Blocks

The NoTeX template provides a clean way to write code snippets in your documents. Everything is built on **minted**, which highlights code automatically through the **pygments** library, so you get proper syntax colouring for free in any of the languages **pygments** knows. The only exception is **pseudocode**, which **pygments** has no lexer for and which therefore gets an environment of its own.

1.1 Inline Code

There are two ways to write code inline. The `\cc` command gives you plain, markdown-style monospaced text with a coloured background and no highlighting at all: `print("Hello, World!")`. The `\inline` command instead runs the text through **minted**, so it comes out highlighted like any other block: `print("Hello, World!")`. The language goes in the optional argument.

1.2 Code Blocks

Every language has an environment named after it. Just open it and write, no arguments needed.

```
<MINTED>
```

```
<MINTED>
```

If the language you need has no environment of its own, use the generic `code` environment and name the language yourself. Any name **pygments** recognises works.

```
<MINTED>
```

1.3 Captions and References

Code blocks are captioned exactly like figures and tables. Add a `caption` key and the block gets a numbered caption underneath; add a `label` key too and you can point at it from anywhere in the text. Numbering follows the section, so the blocks below are ?? and ??, and it resets at every new section.

```
<MINTED>
```

Code 1.1: A function greeting the user by name

A block without a caption is not numbered, just like a figure without one. Math mode works inside comments through the usual `$...$` delimiters.

```
<MINTED>
```

Code 1.2: Squaring a number

Referencing works with every mechanism you would expect: ?? and ?? both resolve, and `\listoflistings` prints a list of every captioned block in the document.

1.4 Pseudocode

Pseudocode gets its own environment. It runs on **listings** rather than **minted**, which buys the one thing pseudocode really needs and **minted** cannot offer: real math mode

anywhere in the body, not only inside comments. Keywords common to pseudocode are automatically highlighted with some predefined words.

```
function shuffle(A):
    // Fisher-Yates shuffle
    N ← length(A)
    for i = 0 to N-1 do
        j ← random integer between 0 and N - 1
        swap A[i] and A[j]
    return A
```

Code 1.3: Shuffling a list in place

You can also write pseudocode with line numbers. By default, line numbers are hidden, but you can show them with the `linenos = true` key as in the following example:

```
1 if x > 0 then
2     return  $\sqrt{x}$ 
3 else
4     error "negative input"
```

1.5 Options

All the code environments take the same optional keys, which you can mix freely.

Key	What it does
<code>caption</code>	Adds a numbered caption below the block
<code>label</code>	Makes the block referenceable with <code>\ref</code> or <code>\cref</code>
<code>linenos</code>	Shows line numbers (<code>linenos=false</code> hides them)
<code>options</code>	Extra raw minted options, or listings options for pseudocode
<code>tcb</code>	Extra raw tcolorbox options, to restyle the box itself

Table 1.1: The optional keys accepted by every code environment

The last two are escape hatches: anything minted or tcolorbox can do stays reachable without touching the template.

```
<MINTED>
```

Code 1.4: A block starting its line numbers at 10

If you want a setting applied to every block in the document, set it once in the preamble with `\notexcodeset`, which takes the same keys. For instance `\notexcodeset{linenos}` numbers every code block you write.

i The opening bracket of the optional argument must sit on the same line as `\begin{...}`, with no space in between, and nothing else may follow the environment name on that line. This is what lets a block start with a character LaTeX would normally choke on, like the `#include` in the C++ example above.